



NED University of Engineering & Technology
Department of Electrical Engineering

Name: **Muhammad Taha** Roll no.: **EE-23319**

Course Code: **EE-354**

Course Title: **Embedded Systems**

Laboratory Session No.: **08**

Date: _____

Psychomotor Domain Assessment Rubric for Laboratory (Level P3)					
Skill(s) to be assessed	Extent of Achievement				
	0	1	2	3	4
Software Initialisation and Configuration: Set up and <u>recognise</u> software initialisation and configuration steps	Completely unable to recognise initialisation and configuration	Able to recognise initialisation but could not configure	Able to recognise initialisation but configuration is erroneous	Able to recognise initialisation and configuration with minimal errors	Able to recognise initialisation and configuration with complete success
Equipment Identification and Handling: <u>Sensory</u> skill to identify equipment and its components along with adherence to safe handling	Completely unable to identify equipment and components and no regard to safe handling		Ability to identify equipment but makes mistakes in recognising components, demonstrates decent equipment handling capacity		Ability to identify equipment and recognises all components, practices careful and safe handling
Establish and Verify Hardware-Software Connection: <u>Recognise</u> interface between computer and hardware kit and <u>establish</u> connectivity with software	Unable to perform hardware and software connection verification		Able to verify hardware connection but unable to establish software connection verification		Able to verify hardware connection and successfully establishes software connection verification
Following step-by-step procedure to complete lab work: <u>Observe</u> , <u>imitate</u> and <u>operate</u> hardware in conjunction with software to complete the provided sequence of steps	Inability to recognise and perform given lab procedures	Able to recognise given lab procedures and perform them but could not follow the prescribed order of steps	Able to recognise given lab procedures and perform them by following prescribed order of steps, with frequent mistakes	Able to recognise given lab procedures and perform them by following prescribed order of steps, with occasional mistakes	Able to recognise given lab procedures and perform them by following prescribed order of steps, with no mistakes

Psychomotor Domain Assessment Rubric for Laboratory (Level P3)					
Skill(s) to be assessed	Extent of Achievement				
	0	1	2	3	4
Programming the Controller for given Embedded System Problem: <u>Imitate</u> and <u>practice</u> given embedded C instructions for implementing specific control strategy and store required variables	Incorrect selection and use of programming constructs and instructions	Correct selection of programming constructs and instructions but their use is incorrect	Correct selection and use of programming constructs and instructions with many syntax/logical errors	Correct selection and use of programming constructs and instructions with little to no syntax/logical errors	Correct selection and use of programming constructs and instructions with no syntax/logical errors
Software Menu Identification and Usage: Ability to <u>operate</u> software environment <u>under supervision</u> , using menus, shortcuts, instructions etc.	Unable to understand and use software menu	Little ability and understanding of software menu operation, makes many mistake	Moderate ability and understanding of software menu operation, makes lesser mistakes	Reasonable understanding of software menu operation, makes no major mistakes	Demonstrates command over software menu usage with occasional use of advance menu options
Detecting and Removing Errors/Exceptions in Hardware and Software: <u>Detect</u> Errors/Exceptions and <u>manipulate</u> , under supervision, to rectify the embedded C program	Unable to check and detect error messages in software and hardware	Able to find error messages in software but no sense of hardware error identification	Able to find error messages in software and recognise them on hardware. Still unable to understand the error type and possible causes	Able to find error messages in software and recognise them on hardware. Moderately able in understanding error type and possible causes	Able to find error messages in software and recognise them on hardware. Reasonably able in understanding error type and possible causes
Visualisation, Comparison and analysis of results: <u>Copy</u> or enter results in analysis software to visualise and compare them with inputs. Use analysis tools to compute standard indices from result	Unable to understand and utilise visualisation, plotting and analysis software	Ability to understand and utilise visualisation and plotting instructions with errors. Unable to compute standard indices	Ability to understand and utilise visualisation and plotting instructions with occasional errors. Able to partially compute standard indices	Ability to understand and utilise visualisation and plotting instructions with no errors. Able to partially compute standard indices	Ability to understand and utilise visualisation and plotting instructions without errors. Able to compute standard indices completely

NED University of Engineering & Technology
Department of Electrical Engineering



Name: **Ali Nasir** Roll no.: **EE-23328**

Course Code: **EE-354**

Course Title: **Embedded Systems**

Laboratory Session No.: **08**

Date: _____

Psychomotor Domain Assessment Rubric for Laboratory (Level P3)					
Skill(s) to be assessed	Extent of Achievement				
	0	1	2	3	4
Software Initialisation and Configuration: Set up and <u>recognise</u> software initialisation and configuration steps	Completely unable to recognise initialisation and configuration	Able to recognise initialisation but could not configure	Able to recognise initialisation but configuration is erroneous	Able to recognise initialisation and configuration with minimal errors	Able to recognise initialisation and configuration with complete success
Equipment Identification and Handling: <u>Sensory</u> skill to identify equipment and its components along with adherence to safe handling	Completely unable to identify equipment and components and no regard to safe handling		Ability to identify equipment but makes mistakes in recognising components, demonstrates decent equipment handling capacity		Ability to identify equipment and recognises all components, practices careful and safe handling
Establish and Verify Hardware-Software Connection: <u>Recognise</u> interface between computer and hardware kit and <u>establish</u> connectivity with software	Unable to perform hardware and software connection verification		Able to verify hardware connection but unable to establish software connection verification		Able to verify hardware connection and successfully establishes software connection verification
Following step-by-step procedure to complete lab work: <u>Observe</u> , <u>imitate</u> and <u>operate</u> hardware in conjunction with software to complete the provided sequence of steps	Inability to recognise and perform given lab procedures	Able to recognise given lab procedures and perform them but could not follow the prescribed order of steps	Able to recognise given lab procedures and perform them by following prescribed order of steps, with frequent mistakes	Able to recognise given lab procedures and perform them by following prescribed order of steps, with occasional mistakes	Able to recognise given lab procedures and perform them by following prescribed order of steps, with no mistakes

Psychomotor Domain Assessment Rubric for Laboratory (Level P3)					
Skill(s) to be assessed	Extent of Achievement				
	0	1	2	3	4
Programming the Controller for given Embedded System Problem: <u>Imitate</u> and <u>practice</u> given embedded C instructions for implementing specific control strategy and store required variables	Incorrect selection and use of programming constructs and instructions	Correct selection of programming constructs and instructions but their use is incorrect	Correct selection and use of programming constructs and instructions with many syntax/logical errors	Correct selection and use of programming constructs and instructions with little to no syntax/logical errors	Correct selection and use of programming constructs and instructions with no syntax/logical errors
Software Menu Identification and Usage: Ability to <u>operate</u> software environment <u>under supervision</u> , using menus, shortcuts, instructions etc.	Unable to understand and use software menu	Little ability and understanding of software menu operation, makes many mistake	Moderate ability and understanding of software menu operation, makes lesser mistakes	Reasonable understanding of software menu operation, makes no major mistakes	Demonstrates command over software menu usage with occasional use of advance menu options
Detecting and Removing Errors/Exceptions in Hardware and Software: <u>Detect</u> Errors/Exceptions and <u>manipulate</u> , under supervision, to rectify the embedded C program	Unable to check and detect error messages in software and hardware	Able to find error messages in software but no sense of hardware error identification	Able to find error messages in software and recognise them on hardware. Still unable to understand the error type and possible causes	Able to find error messages in software and recognise them on hardware. Moderately able in understanding error type and possible causes	Able to find error messages in software and recognise them on hardware. Reasonably able in understanding error type and possible causes
Visualisation, Comparison and analysis of results: <u>Copy</u> or enter results in analysis software to visualise and compare them with inputs. Use analysis tools to compute standard indices from result	Unable to understand and utilise visualisation, plotting and analysis software	Ability to understand and utilise visualisation and plotting instructions with errors. Unable to compute standard indices	Ability to understand and utilise visualisation and plotting instructions with occasional errors. Able to partially compute standard indices	Ability to understand and utilise visualisation and plotting instructions with no errors. Able to partially compute standard indices	Ability to understand and utilise visualisation and plotting instructions without errors. Able to compute standard indices completely